

API Specification Doc

Healthplix EMR - Hospital integration

Version	Date	Author	Description
1.0	13-June-2023	Mohammed Akzin K	Initial draft
1.1	27-July-2023	Mohammed Akzin K	Added APIs for Appointment Service, Doctor & Services Listing
1.2	19-Dec-2023	Mohammed Akzin K	Added APIs for Patient details edit/update.
1.3	07-Mar-2024	Mohammed Akzin K	Updated APIs with sample request body, additional header
1.4	07-May-2024	Srinivasa Raghavan Kandala	Added API for combined operation of Patient and Appointment creation
1.5	21-Jun-2024	Srinivasa Raghavan Kandala	Order of APIs listed is changed.
1.6	29-Jul-2024	Srinivasa Raghavan Kandala	URL correction (removed trailing slash) Index updated
1.7	26-Apr-2025	Srinivasa Raghavan Kandala	Added API for saving vitals & fetching vital test identifiers

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APIs

Base URL for QA environment: <https://consultation-edge-dev.healthplix.com/>

1. Generate Token

Authenticate the tenant with the system and obtain the access token

Request

Method	URL
POST	v1/generate/token

Headers

Parameter Name	Description
Basic Auth - username*	tenant_id which was provided during registration
Basic Auth - password*	tenant_secret which was provided during registration

Note: fields marked * are mandatory.

Response

Status	Response
200	{ "access_token": <jwt_token> , "expiry": <int>

	<pre>} access_token (string) - all further API calls must have this key in header as bearer token expiry (int) - expiry of the token in minutes</pre>
400	<pre>{"error": "Please provide username."}</pre>
400	<pre>{"error": "Please provide password."}</pre>
401	<pre>{"error": "Invalid Tenant Id."}</pre>
401	<pre>{"error": "Invalid Tenant Secret."}</pre>
401	<pre>{"error": "Incorrect Tenant details, please pass valid tenant details."}</pre>
500	<pre>{"error": "Something went wrong. Please try again later."}</pre>

2. Register Patient And Appointment

To register a new patient and appointment onto the Healthplix EMR.

Note: If there is more than one patient record found for the given patient_identifier, the API will not update the details.

Request

Method	URL
POST	v1/appointment-and-patient/register

Headers

Parameter Name	Description
Bearer Token*	access_token that is generated using the v1/token/generate api

tenant_id*	tenant_id which was provided during registration
content-type*	application/json

Parameters

Parameter Name	Type	Description
patient_details*	object	Patient details with same parameters as the register patient API
appointment_details*	object	Appointment details with same parameters as the register appointment API

Note: fields marked * are mandatory.

Sample Request Body

```
{
  "patient_details": {
    "patient_identifier": "some_pat_identifier",
    "dob": "1970-01-01",
    "name": "Demo Patient Edit",
    "gender": "male",
    "honorific": "mr",
    "email": "demo_patient@email.com",
    "address": "demo room, demo building",
    "city": "demo city",
    "pincode": "1111111",
    "phone_secondary": "8888899999",
    "occupation": "demo occupation",
    "tag": "demo_tag"
  },
  "appointment_details": {
    "patient_identifier": "some_pat_identifier",
    "doctor_identifier": "some_doc_identifier",
    "appnt_date": "2023-12-28",
    "appnt_time": "09:00:42",
    "appnt_duration": 30,
    "service_identifier": "some_serv_identifier"
  }
}
```

Response

Status	Response
202	Response will be an object status of the request. Once a patient is added into the database, a webhook will be invoked with patient details. <pre>{ "status": true, "message": "Patient and Appointment registration request has been accepted." }</pre>
400	<pre>{"error": "Invalid patient_identifler/name/gender/honorific/dob"}</pre>
400	<pre>{"error": "Invalid dob format."}</pre>
401	<pre>{"error": "Invalid access token."}</pre>
401	<pre>{"error": "Access token expired."}</pre>
403	<pre>{"error": "Access token expired."}</pre>
500	<pre>{"error": "Something went wrong. Please try again later."}</pre>

Webhook

Event Name	Response
appointment_and_patient.registered	Triggered when appointment and patient are registered successfully into the Healthplix EMR
patient.rejected	Triggered when patient registration is rejected due to some reason
patient.failed	Triggered when patient registration failed

appointment.rejected	Triggered when appointment registration is rejected due to some reason
appointment.failed	Triggered when appointment registration failed

Appointment and Patient Registered

```
{
  "entity": "event",
  "event": "appointment_and_patient.registered",
  "payload": {
    "patient": {
      "id": "123",
      "entity": "patient"
    },
    "status_details": {
      "description": "Patient And Appointment has been registered
in Healthplix EMR",
      "source": "healthplix_emr",
      "reason": "appointment_and_patient_registered"
    }
  },
  "created_at": 1580808301
}
```

Patient Rejected

```
{
  "entity": "event",
  "event": "patient.rejected",
  "payload": {
    "patient": {
      "id": "123",
      "entity": "patient"
    },
    "status_details": {
      "description": "Patient registration has been rejected
because of a conflict in reason_for_rejection",
      "source": "healthplix_emr",
      "reason": "patient_rejected"
    }
  }
}
```

```
    }  
  },  
  "created_at":1580808301  
}
```

Reason_for_rejection values:

- patient id

Patient Failed

```
{  
  "entity":"event",  
  "event":"patient.failed",  
  "payload":{  
    "patient":{  
      "id":"123",  
      "entity":"patient"  
    },  
    "status_details": {  
      "description": "Patient registration has failed because of  
some reason. Please try again",  
      "source": "healthplix_emr",  
      "reason": "patient_failed"  
    }  
  },  
  "created_at":1580808301  
}
```

Appointment Rejected

```
{  
  "entity":"event",  
  "event":"appointment.rejected",  
  "payload":{  
    "status_details": {  
      "description": "Patient has been registered into Healthplix  
EMR, Appointment registration has been rejected : reason_for_rejection",  
      "source": "healthplix_emr",  
      "reason": "appointment_rejected",  
    }  
  }  
}
```



```

        "req_id": <unique_request_id>,
      },
      "created_at":1580808301
    }
  }
}

```

Reason_for_rejection values:

- invalid patient identifier
- invalid service identifier
- Something went wrong ! - for any other errors
- invalid doctor identifier
- cannot create appointment for a past date

Appointment Failed

```

{
  "entity":"event",
  "event":"appointment.failed",
  "payload":{
    "status_details": {
      "description": "Patient has been registered into Healthplix
EMR, Appointment registration has failed because of some reason. Please try
again",
      "source": "healthplix_emr",
      "reason": "appointment_failed",
      "req_id": <unique_request_id>,
    }
  },
  "Created_at":1580808301
}

```

3. Register Patient

(Please use this if the Patient and Appointment registration are independent flows in the HIS system)

To register a new patient onto the Healthplix EMR

Request

Method	URL
POST	v1/patient/register

Headers

Parameter Name	Description
Bearer Token*	access_token that generated using the v1/token/generate api
tenant_id*	tenant_id which was provided during registration
content-type*	type/xml or application/json

Parameters

Parameter Name	Type	Description
patient_identifier*	string	The unique identifier of the patient in the HIS system. Identifier will be case insensitive and should contain minimum of 3 characters

name*	string	The full name of the patient
phone*	string	The primary contact number of the patient
gender*	string	The gender of the patient. Valid values are (case insensitive) 'M' or "Male" 'F' or "Female" 'O' or "Others"
honorific*	string	The honorific of the patient. Eg Mr/Mrs/Ms etc
dob*	string	The date of birth of the patient Format: YYYY-MM-DD. Eg: 2023-06-12
email	string	The email id of the patient
address	string	The address of the patient
city	string	The city of the patient
pincode	string	The pincode of the patient
phone_secondary	string	The secondary contact number of the patient
occupation	string	The occupation of the patient
tag	string	Comma separated tags that can be applied to a patient

Note: fields marked * are mandatory.

Sample Request Body

```
{
  "name": "Demo Patient",
  "phone": "9999999999",
  "gender": "M",
  "honorific": "Mr.",
  "patient_identifier": "some_pat_identifier"
  "dob": "1970-01-01"
}
```

Response

Status	Response
202	Response will be an object status of the request. Once a patient is added into the database, a webhook will be invoked with patient details. <pre>{ "status": true, "message": "Patient registration request has been accepted." }</pre>
400	<pre>{"error": "patient_identifier/name/gender/honorific/dob cannot be blank"}</pre>
400	<pre>{"error": "Invalid patient_identifier/name/gender/honorific/dob"}</pre>
400	<pre>{"error": "Invalid dob format."}</pre>
401	<pre>{"error": "Invalid access token."}</pre>
401	<pre>{"error": "Access token expired."}</pre>
403	<pre>{"error": "Access token expired."}</pre>
500	<pre>{"error": "Something went wrong. Please try again later."}</pre>

Webhook

Event Name	Response
patient.registered	Triggered when patient is registered successfully into the Healthplix EMR

patient.rejected	Triggered when patient registration is rejected due to some reason
patient.failed	Triggered when patient registration failed

Patient Registered

```
{
  "entity": "event",
  "event": "patient.registered",
  "payload": {
    "patient": {
      "id": "123",
      "entity": "patient"
    },
    "status_details": {
      "description": "Patient has been registered into Healthplix
EMR",
      "source": "healthplix_emr",
      "reason": "patient_registered"
    }
  },
  "created_at": 1580808301
}
```

Patient Rejected

```
{
  "entity": "event",
  "event": "patient.rejected",
  "payload": {
    "patient": {
      "id": "123",
      "entity": "patient"
    },
    "status_details": {
      "description": "Patient registration has been rejected
because of a conflict in reason_for_rejection",
      "source": "healthplix_emr",

```

```
        "reason": "patient_rejected"
      }
    },
    "created_at":1580808301
  }
}
```

Reason_for_rejection values:

- patient id

Patient Failed

```
{
  "entity":"event",
  "event":"patient.failed",
  "payload":{
    "patient":{
      "id":"123",
      "entity":"patient"
    },
    "status_details": {
      "description": "Patient registration has failed because of
some reason. Please try again",
      "source": "healthplix_emr",
      "reason": "patient_failed"
    }
  },
  "created_at":1580808301
}
```

4. Register Appointment

(Please use this if the Patient and Appointment registration are independent flows in the HIS system)

To register a new appointment onto the Healthplix EMR

Request

Method	URL
POST	/v1/appointment/register

Headers

Parameter Name	Description
Bearer Token*	access_token that generated using the v1/token/generate api
tenant_id*	tenant_id which was provided during registration
content-type*	application/json

Parameters

Parameter Name	Type	Description
patient_identifier*	string	The unique identifier of the patient in the HIS system. Identifier will be case insensitive and should contain minimum of 3 characters
doctor_identifier*	string	The unique identifier of the doctor ID shared by HealthPlix

appnt_time*	string	The appointment time Format: HH:MM:SS Eg. 15:04:05
appnt_date*	string	The appointment date Format: YYYY-MM-DD Eg: 2023-06-12
appnt_duration*	int	The duration of the appointment (in minutes)
service_identififier*	string	The unique identifier of the service id shared by HealthPlix

Note: fields marked * are mandatory.

Sample Request Body

```
{
  "patient_identifier": "some_pat_identifier",
  "doctor_identifier": "some_doc_identifier",
  "appnt_date": "2023-12-28",
  "appnt_time": "09:00:42",
  "appnt_duration": 30,
  "service_identififier": "some_serv_identifier"
}
```

Response

Status	Response
202	Response will be an object status of the request. Once an appointment is added into the database, a webhook will be invoked with appointment details. <pre>{ "status": true, "message": "Appointment registration request has been accepted.", "req_id": <unique_request_id> }</pre>
400	<pre>{"error": "patient_identifier/doctor_identifier/appnt_durati</pre>

	on/service_identifier cannot be blank"} }
400	{"error": "invalid doctor_identifier"}
400	{"error": "Invalid appt_time/appt_date format."}
401	{"error": "Invalid access token."}
401	{"error": "Access token expired."}
500	{"error": "Something went wrong. Please try again later."}

Webhook

Event Name	Response
appointment.registered	Triggered when appointment is registered successfully into the Healthplix EMR
appointment.rejected	Triggered when appointment registration is rejected due to some reason
appointment.failed	Triggered when appointment registration failed

Appointment Registered

```
{
  "entity": "event",
  "event": "appointment.registered",
  "payload": {
    "status_details": {
      "description": "Appointment has been registered into
Healthplix EMR",
      "source": "healthplix_emr",
      "reason": "appointment_registered",
      "req_id": <unique_request_id>,
    }
  },
  "created_at": 1580808301
}
```

```
}
```

Appointment Rejected

```
{
  "entity": "event",
  "event": "appointment.rejected",
  "payload": {
    "status_details": {
      "description": "Appointment registration has been rejected :
reason_for_rejection",
      "source": "healthplix_emr",
      "reason": "appointment_rejected",
      "req_id": <unique_request_id>,
    }
  },
  "created_at": 1580808301
}
```

Reason_for_rejection values:

- invalid patient identifier
- invalid service identifier
- Something went wrong ! - for any other errors
- invalid doctor identifier
- cannot create appointment for a past date

Appointment Failed

```
{
  "entity": "event",
  "event": "appointment.failed",
  "payload": {
    "status_details": {
```

```

        "description": "appointment registration has failed because
of some reason. Please try again",
        "source": "healthplix_emr",
        "reason": "appointment_failed",
        "req_id": <unique_request_id>,
    }
},
"Created_at":1580808301
}

```

5. Fetch Doctors

Fetch list of doctors available at branch

Request

Method	URL
GET	v1/doctors

Headers

Parameter Name	Description
Bearer Token*	access_token that generated using the v1/token/generate api
tenant_id*	tenant_id which was provided during registration

Note: fields marked * are mandatory.

Response

Status	Response
--------	----------

200	<pre>{ "doctor_list": [{ doctor_name : <string>, doctor_identifier: <string>, doctor_phone : <string>, doctor_email: <string>, doctor_speciality: <string> }] , "total_count": <int> }</pre>
401	<pre>{"error": "Access token expired."}</pre>
401	<pre>{"error": "Invalid access token."}</pre>
500	<pre>{"error": "Something went wrong. Please try again later."}</pre>

6. Fetch Services

Fetch list of services available at branch

Request

Method	URL
GET	v1/services

Headers

Parameter Name	Description
Bearer Token*	access_token that generated using the v1/token/generate api
tenant_id*	tenant_id which was provided during registration

Note: fields marked * are mandatory.

Response

Status	Response
200	<pre>{ "service_list": [{ service_name : <string>, service_identifier: <string>, service_owner: <string>, price: <string>, service_category: <string>, }] , "total_count": <int> }</pre>
401	{"error": "Invalid access token."}
401	{"error": "Access token expired."}
500	{"error": "Something went wrong. Please try again later."}

7. Fetch Doctors and Services

Fetch list of doctors and services available at branch

Request

Method	URL
GET	v1/doctors-and-services

Headers

Parameter Name	Description
Bearer Token*	access_token that generated using the v1/token/generate api
tenant_id*	tenant_id which was provided during registration

Note: fields marked * are mandatory.

Response

Status	Response
200	<pre>{ "service_list": [{ service_name : <string>, service_identifier: <string>, service_owner: <string>, price: <string>, service_category: <string> }] , "doctor_list": [{ doctor_name : <string>, doctor_identifier: <string>, doctor_phone : <string>, doctor_email: <string>, doctor_speciality: <string> }] , "total_service_count": <int>, "total_doctor_count": <int> }</pre>
401	<pre>{"error": "Invalid access token."}</pre>
401	<pre>{"error": "Access token expired."}</pre>
500	<pre>{"error": "Something went wrong. Please try again later."}</pre>

8. Update Patient Details

To update patient details in the Healthplix EMR.

Note: If there is more than one patient record found for the given patient_identifier, the API will not update the details.

Request

Method	URL
POST	v1/patient/update

Headers

Parameter Name	Description
Bearer Token*	access_token that generated using the v1/token/generate api
tenant_id*	tenant_id which was provided during registration
content-type*	application/json

Parameters

Parameter Name	Type	Description
patient_identifier*	string	The unique identifier of the patient in the HIS system. Identifier will be case insensitive and should contain minimum of 3 characters
name*	string	The full name of the patient

phone*	string	The primary contact number of the patient
gender*	string	The gender of the patient. Valid values are (case insensitive) 'M' or "Male" 'F' or "Female" 'O' or "Others"
honorific*	string	The honorific of the patient. Eg Mr/Mrs/Ms etc
dob*	string	The date of birth of the patient Format: YYYY-MM-DD. Eg: 2023-06-12
email	string	The email id of the patient
address	string	The address of the patient
city	string	The city of the patient
pincode	string	The pincode of the patient
phone_secondary	string	The secondary contact number of the patient
occupation	string	The occupation of the patient
tag	string	Comma separated tags that can be applied to a patient

Note: fields marked * are mandatory.

Sample Request Body

```
{
  "patient_identifier": "some_pat_identifier",
  "dob": "1970-01-01",
  "name": "Demo Patient Edit",
  "gender": "male",
  "honorific": "mr",
  "email": "demo_patient@email.com",
  "address": "demo room, demo building",
```



```
"city": "demo city",
"pincode": "1111111",
"phone_secondary": "8888899999",
"occupation": "demo occupation",
"tag": "demo_tag"
}
```

Response

Status	Response
202	Response will be an object status of the request. Once a patient is added into the database, a webhook will be invoked with patient details. <pre>{ "status": true, "message": "Patient details updation request has been accepted." }</pre>
400	<pre>{"error": "patient_identifier"}</pre>
400	<pre>{"error": "Invalid patient_identifier/name/gender/honorific/dob"}</pre>
400	<pre>{"error": "Invalid dob format."}</pre>
401	<pre>{"error": "Invalid access token."}</pre>
401	<pre>{"error": "Access token expired."}</pre>
403	<pre>{"error": "Access token expired."}</pre>
500	<pre>{"error": "Something went wrong. Please try again later."}</pre>

Webhook

Event Name	Response
patient.updated	Triggered when patient is updated successfully into the Healthplix EMR
patient.rejected	Triggered when patient registration is rejected due to some

	reason
patient.failed	Triggered when patient registration failed

Patient Updated

```
{
  "entity": "event",
  "event": "patient.updated",
  "payload": {
    "patient": {
      "id": "123",
      "entity": "patient"
    },
    "status_details": {
      "description": "Patient details have been updated in
Healthplix EMR",
      "source": "healthplix_emr",
      "reason": "patient_updated"
    }
  },
  "created_at": 1580808301
}
```

Patient Rejected

```
{
  "entity": "event",
  "event": "patient.rejected",
  "payload": {
    "patient": {
      "id": "123",
      "entity": "patient"
    },
    "status_details": {
      "description": "Patient details updation has been rejected
because reason_for_rejection",
      "source": "healthplix_emr",

```

```

        "reason": "patient_update_rejected"
      }
    },
    "created_at": 1580808301
  }
}

```

Reason_for_rejection values:

- patient id

Patient Failed

```

{
  "entity": "event",
  "event": "patient.failed",
  "payload": {
    "patient": {
      "id": "123",
      "entity": "patient"
    },
    "status_details": {
      "description": "Patient details updation has failed because
of some reason. Please try again",
      "source": "healthplix_emr",
      "reason": "patient_update_failed"
    }
  },
  "created_at": 1580808301
}

```

9. Record Vitals

To record vitals of a patient in the Healthplix EMR.

Note: If there is more than one patient record found for the given patient_identifier, the API will not record the details.

Request

Method	URL
POST	v1/tests

Headers

Parameter Name	Description
Bearer Token*	access_token that generated using the v1/token/generate api
tenant_id*	tenant_id which was provided during registration
content-type*	application/json

Parameters

Parameter Name	Type	Description
patient_identifier*	string	The unique identifier of the patient in the HIS system. Identifier will be case insensitive and should contain minimum of 3 characters
doctor_identifier*	string	The unique identifier of the doctor (shared by Healthplix)
tests.test_identifier*	string	The unique identifier of the Vital / Test (shared by Healthplix). The test_identifier is received in the “Fetch Vitals Test” API.
tests.test_date*	string	Date of test in YYYY-mm-dd format
tests.test_value*	string	The value of test
tests.out_of_range	boolean	Determine if the test value falls outside the permitted range. Pass

		the value based on the permitted range of value for this test as per your system if any.
--	--	--

Note: fields marked * are mandatory.

Sample Request Body

<pre>{ "patient_identifier": "pat_identifier", "doctor_identifier": "doctor_identifier", "tests": [{ "test_identifier": "test_identifier", "test_date": "2025-04-22", "test_value": "20", "out_of_range": true }, { "test_identifier": "test_identifier", "test_date": "2025-04-22", "test_value": "20", "out_of_range": true }] }</pre>
--

Response

Status	Response
202	Response will be an object status of the request. Once a test is added into the database, a webhook will be invoked with patient details. <pre>{ "status": true, "message": "Test details save request has been accepted." }</pre>
400	{"error": "patient_identifier"}
400	{"error": "Invalid patient_identifier"}
400	{"error": "Invalid test_date format."}
401	{"error": "Invalid access token."}

401	{"error": "Access token expired."}
403	{"error": "Access token expired."}
500	{"error": "Something went wrong. Please try again later."}

Webhook

Event Name	Response
test.saved	Triggered when vitals are saved successfully into the Healthplix EMR
test.failed	Triggered when vitals save has failed

Test Saved

```
{
  "entity": "event",
  "event": "test.saved",
  "payload": {
    "status_details": {
      "description": "Test details has been saved into Healthplix EMR",
      "source": "healthplix_emr",
      "reason": "test_saved",
      "reqId": "your-request-id"
    }
  },
  "created_at": 1580808301
}
```

Test Rejected

```
{
  "entity": "event",
  "event": "test.failed",
  "payload": {
    "status_details": {
      "description": "Test save has failed because of some reason.
Please try again",
      "source": "healthplix_emr",
      "reason": "test_save_failed",
      "reqId" : "your-request-id"
    }
  },
  "created_at": 1580808301
}
```

10. Fetch Vital tests

Fetch list of vital test identifiers available at branch

Request

Method	URL
GET	v1/tests?fetch_vitals=true

Headers

Parameter Name	Description
Bearer Token*	access_token that generated using the v1/token/generate api
tenant_id*	tenant_id which was provided during registration

Note: fields marked * are mandatory.

Response

Status	Response
200	<pre>{ "test_list": [{ "test_identifier" : <string>, "test_name": <string>, "test_unit": <string>, "test_range_text": <string>, "test_range_rule": <string>, "category": <string> }], "total_count": <int> }</pre>
401	<pre>{"error": "Invalid access token."}</pre>
401	<pre>{"error": "Access token expired."}</pre>
500	<pre>{"error": "Something went wrong. Please try again later."}</pre>

Glossary

Conventions

- **Client** - Client application.
- **Status** - HTTP status code of response.
- **Webhook** - Client webhook which will be used to send responses for async processes
- All the possible responses are listed under 'Responses' for each method. Only one of them is issued per request server.
- All requests and responses are in JSON format.
- All request parameters marked as * are mandatory

Status Codes

All status codes are standard HTTP status codes. The below ones are used in this API.

2XX - Success of some kind

4XX - Error occurred in client's part

5XX - Error occurred in server's part

Status Code	Description
200	OK
201	Created
202	Accepted (Request accepted, and queued for execution)
400	Bad request
401	Authentication failure
403	Forbidden
404	Resource not found
405	Method Not Allowed
409	Conflict
413	Request Entity Too Large
500	Internal Server Error

501	Not Implemented
503	Service Unavailable